

CATALIN URSU

UNITY GAMEPLAY ENGINEER | UNITY DEVELOPER | GAMEPLAY PROGRAMMER

Helsinki, Finland · Remote only · EU citizen — no visa sponsorship required

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SUMMARY

Unity Gameplay Engineer with 6+ years of commercial experience building gameplay systems, scalable architecture and UI for mobile (iOS/Android), AR, VR and PC. Shipped 12+ production projects for clients including NASA, Deloitte, PwC and Mercedes.

Raised frame rate from ~40 to ~70 FPS and cut memory usage by 30% on production builds. Strong background in dependency injection (Zenject), async and reactive workflows (UniTask, R3), and profiling-driven optimization.

SKILLS

Core: Unity (2021 LTS – Unity 6), C#, OOP, Gameplay Systems, Mobile (iOS, Android), VR/AR, PC

Architecture & Async: SOLID, DRY, Dependency Injection (Zenject), Event-driven architecture, Design patterns, UniTask, async/await, R3 (ReactiveProperty)

Performance & Graphics: Unity Profiler, Frame Debugger, Memory Profiler, CPU/GPU optimization, Draw call batching, URP, Shaders, VFX, Render Textures, Cinemachine

UI & Assets: UGUI, TextMeshPro, DoTween, Adaptive layouts, Safe Area, Localization, Addressables, Asset optimization, Resource management

Multiplayer & Audio: Netcode for GameObjects, Client-server synchronization, Cross-platform MR, FMOD

Workflow: Git, GitHub, Code review, Agile, Scrum, Jira

WORK EXPERIENCE

Bully Entertainment — Unity Gameplay Engineer

Chisinau, Moldova · Remote (from Helsinki since 2021) | Apr 2021 – Present

- Raised frame rate from ~40 to ~70 FPS by profiling CPU/GPU load and removing bottlenecks with Unity Profiler, Frame Debugger and Memory Profiler
- Cut memory usage 30% (1.08 GB → 0.88 GB) through Addressables, asset compression and tighter resource management
- Architected modular gameplay systems with Zenject (DI) and UniTask, shortening feature delivery time across 12+ client projects
- Built and continually reworked a resolution-independent UI system with Safe Area support (4:3–21:9), extending it across new designs and mobile devices without regressions
- Rebuilt UI interaction logic on R3, replacing per-frame Update loops with reactive flows and removing avoidable allocations
- Shipped a DLC system with card-based mechanics, including deck generation logic with guaranteed win conditions
- Delivered multiplayer synchronization for a cross-platform MR project (VR + PC), keeping shared world state consistent across devices

Freelance Unity Developer

Chisinau, Moldova (remote) | Sep 2020 – Mar 2021

- Delivered contract Unity work and self-directed projects focused on gameplay systems and reusable architecture
- Developed the foundation of open-source Unity packages later published as Helpers and IdleNumber

Midnight Works — Unity Developer (Full-time)

Chisinau, Moldova | Jun 2020 – Sep 2020

- Developed gameplay prototypes for 2D and 3D mobile projects
- Implemented core gameplay mechanics, UI systems and monetization integrations (AdMob, Facebook SDK)
- Optimized performance for low-end mobile devices

PROJECTS

Showcase — 2D game, architecture reference 2023 – Present github.com/CatalinUrsu/Showcase

- Built a complete 2D game as a reference architecture using Zenject (DI), UniTask and R3
- Implemented reactive, event-driven UI interaction logic with no Update-based polling

Helpers — reusable Unity package 2024 – Present github.com/CatalinUrsu/Tool_Helpers

- Published a modular Unity package (git-URL install) covering FMOD audio, object pooling, a save system, an async state machine (UniTask), and camera/scene-loading services
- Extracted modules from production projects so each system works standalone or combined

IdleNumber — custom numeric type for idle games 2024 github.com/CatalinUrsu/Tool_IdleNumber

- Implemented an abbreviated number type (K, M, B through Q, then AA, AB, AC...) with fixed or dynamic decimal precision
- Tuned the conversion algorithm for low CPU and memory overhead during real-time UI updates

EDUCATION

State University of Moldova — Chisinau, Moldova | 2017 – 2020

BSc, Computer Science — Faculty of Mathematics and Computer Science

LANGUAGES

Romanian — Native | Russian — Native | English — B2 (Upper Intermediate)